

Real-Time Cyber Attack Detection in Power Systems

A PROJECT REPORT

submitted by

CB.EN.U4ELC21007

Arman Mahapatra

CB.EN.U4ELC21054

Sathvika Venigandla

CB.EN.U4ELC21062

Suneelvijay M

CB.EN.U4ELC21064

T. S. Ramsubramania Raja

*in partial fulfillment for the award of the degree
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AMRITA VISHWA VIDYAPEETHAM

COIMBATORE- 641112

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AMRITA VISHWA VIDYAPEETHAM
AMRITA SCHOOL OF ENGINEERING, COIMBATORE, 641112



BONAFIDE CERTIFICATE

This is to certify that the project report entitled, **Real-Time Cyber Attack Detection in Power Systems.**

submitted by

CB.EN.U4ELC21007	Arman Mahapatra
CB.EN.U4ELC21054	Sathvika Venigandla
CB.EN.U4ELC21062	Suneelvijay M
CB.EN.U4ELC21064	T. S. Ramsubramania Raja

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in ELECTRICAL & COMPUTER ENGINEERING** is a Bonafide record of the work carried out under my guidance and supervision at Amrita School of Engineering, Coimbatore

sign

Dr. Balamurugan S.
CHAIRPERSON,
Department of Electrical and Electronics
Engineering, Coimbatore.

Dr. Balamurugan S.
CHAIRPERSON,
Department of Electrical and Electronics
Engineering, Coimbatore.

Mr. Nakkeeran M.
Assistant Professor,
Department of Electrical and Electronics
Engineering, Coimbatore.

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This report represents the partial completion of our project, and we look forward to continuing our work towards its full realization.

ABSTRACT

Rapid digitalisation of power systems has introduced new vulnerabilities to cyberattacks and threatened the security, stability, and reliability of critical energy infrastructures. This paper presents a real-time cyberattack detection framework specifically designed for modern electrical networks. The proposed system integrates IoT-based monitoring devices, secure communication protocols, and advanced data analysis to detect and mitigate cyberattacks in real time. A combination of NodeMCU, PZEM004T sensors and Raspberry Pi is used to monitor key electrical grid parameters such as voltage, current, power and frequency. Data are transmitted to the central server using the MQTT protocol, ensuring low latency and secure communication. Machine learning algorithms are used to detect anomalies and prevent attacks such as denial of service (DoS), false data injection (FDI), and man-in-the-middle (MitM). The effectiveness of the system is confirmed by simulations and real-time data analysis, which demonstrate its ability to improve the resilience of the network. Furthermore, this framework is scalable and can be integrated with cloud-based systems for advanced data storage and visualisation. This research makes an important contribution to ongoing efforts to protect power grids against emerging cyber threats and offers robust, efficient, and scale-up solutions to protect critical infrastructures.

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Chapter 1

INTRODUCTION

1.1 PREFACE

This project focuses on the development of an AI enhanced decision support system. In an era where power systems are becoming increasingly digitized and interconnected, the need for robust cybersecurity measures has never been more critical. Power grids are now at risk from a variety of cyber threats, which can result in severe disruptions to essential services and pose significant risks to national security. This project, titled *Real-Time Cyber Attack Detection in Power Systems*, focuses on designing and implementing a comprehensive framework to safeguard these vital infrastructures from emerging cyber threats.

At its core, the project integrates modern technologies such as the Internet of Things (IoT), machine learning, and data analytics into power systems to enable continuous monitoring and real-time detection of cyber-attacks. By using sensor networks and advanced communication protocols, real-time data from the power grid is collected and analyzed to detect anomalies that could signify potential threats, such as Denial of Service (DoS) attacks, false data injections, and other malicious activities.

This project aims to create a responsive and adaptive defense system for power grids by leveraging machine learning models trained on attack simulations and historical data. These models can accurately identify patterns of cyber-attacks, providing early warnings and enabling timely interventions to prevent damage. Furthermore, the project explores scalable solutions for handling large amounts of data while maintaining system efficiency and security.

With the integration of IoT devices, data analytics, and machine learning, this framework offers a proactive approach to enhancing the security, stability, and reliability of modern power grids. By addressing current vulnerabilities, it provides a foundation for future developments in smart grid cybersecurity, contributing to a safer, more resilient energy infrastructure.

1.2 OBJECTIVE

Real-time Monitoring of Power Grid Parameters:

To develop a system capable of continuously monitoring electrical parameters (voltage, current, power consumption) from multiple consumer points in real-time through end nodes and current transformers.

Anomaly Detection Using Machine Learning:

To design and implement a machine learning model that can detect anomalies in power grid data, indicating potential operational issues or cybersecurity threats, such as false data injection (FDI) or abnormal power usage patterns.

Cybersecurity Threat Detection and Response:

To identify and simulate various cyber threats (e.g., Denial of Service (DoS), Replay Attacks, False Data Injection, Switching Attacks) that may target different layers of the system and develop strategies to detect and mitigate them in real time.

Data Integrity and Security:

To ensure the integrity and accuracy of the data being transmitted from the perception layer (end nodes) to the network layer (edge node) and up to the application layer (central data storage and ML models), even under potential cyberattacks.

Mitigation of Cyber Attacks:

To implement and evaluate effective mitigation techniques to counteract threats such as DoS, Replay, and FDI attacks, ensuring minimal disruption to the grid's operations.

Improving Resilience of the Power Grid System:

To enhance the resilience of the smart grid system by incorporating secure communication protocols, fault tolerance mechanisms, and redundancy to minimize the impact of cybersecurity incidents on grid performance.

Efficient Data Aggregation and Communication:

To optimize the wireless data transmission and communication between the perception layer (end nodes) and network layer (edge node) ensuring efficient data aggregation with minimal latency or packet loss.

Alert Generation and Visualization:

To create a real-time alerting system that notifies grid operators immediately when any abnormal behavior or cybersecurity threat is detected, and to provide a user-friendly dashboard for visualizing the current status of the grid.

Validation and Testing of the System:

To rigorously test and validate the system's ability to withstand various cyberattacks while maintaining operational efficiency, ensuring robustness and reliability under normal conditions and during cyber threats.

Deployment of a Scalable Smart Grid Security Framework:

To develop a scalable framework that can be deployed in different smart grid environments, allowing for easy integration of additional nodes, and consumers, and adapting to evolving cybersecurity threats.

Chapter 2

LITERATURE REVIEW

Albasheer et al. (2023) investigate various cyber-attacks targeting smart grids, including Denial of Service (DoS), data manipulation, false data injection and replay attacks. To address these threats, the study explores the use of machine learning algorithms such as Support Vector Machines (SVM), Decision Trees, and Neural Networks for anomaly detection and real-time monitoring. The research also highlights the application of big data analytics to enhance the detection of these attacks by analyzing large volumes of data generated from smart grid systems. The proposed methods were validated using simulations and case studies, showing significant improvements in detecting and mitigating cyber-attacks in smart grids, thereby enhancing the security and reliability of the grid infrastructure. [1]

Hossain et al. (2019) conducted an in-depth review on the application of big data and machine learning in smart grids, emphasizing the security challenges these technologies introduce. The study highlights the use of machine learning algorithms like Support Vector Machines (SVM), Neural Networks, and Deep Learning for tasks such as load forecasting, anomaly detection, and system optimization. The authors identify significant cyber threats, including Denial of Service (DoS), Man-in-the-Middle (MitM), and False Data Injection Attacks (FDIA). To counter these threats, the paper discusses various security measures and machine learning techniques aimed at improving the resilience of smart grids. The review also underscores the need for advanced security frameworks and future research to address the growing complexities of smart grid systems. [2]

Joha et al. (2024) developed an IoT-based smart energy monitoring, management, and protection system for smart microgrids. The system integrates ESP32 and ESP8266 modules with the Blynk IoT cloud platform to provide real-time monitoring, control, and protection of microgrid energy resources. It offers features such as overload protection, power theft detection, and efficient energy management. The system enables remote monitoring and control via smartphones or computers, improving energy efficiency, grid stability, and security in smart microgrids. [3]

Naithani et al. (2024) evaluated several machine learning algorithms for detecting DDoS attacks in cloud environments using the KDD Cup 1999 dataset. The study found that Random Forest and SVM outperformed other models, achieving near-perfect accuracy, with Random Forest reaching 99.99%. These algorithms demonstrated superior precision and recall, making them the most effective for DDoS detection. Other models, such as Naive Bayes and Isolation Forest, showed lower performance, highlighting the importance of selecting the right algorithm for cloud security. [4]

The authors (Kumar et al., 2024) proposed a nature-inspired feature selection method named Sheep Flock Optimization Algorithm-Least Absolute Shrinkage and Selection Operator (SFOA-LASSO) for building a lightweight Intrusion Detection System (IDS) in Software-Defined Internet of Things (SD-IoT) environments. This technique optimizes feature selection by combining the exploration and exploitation capabilities of the Sheep Flock Optimization Algorithm (SFOA) with the regularization properties of LASSO. The SFOA component dynamically adjusts feature selection based on fitness values, while LASSO refines the selection by eliminating redundant features. The performance of the proposed SFOA-LASSO method was evaluated on SD-IoT datasets, demonstrating high detection accuracy and efficiency with minimal computational overhead, making it a strong candidate for securing resource-constrained SD-IoT applications. [5]

The study conducted by Zhang et al. (2023) proposes a novel detection method for False Data Injection Attacks (FDIAs) in smart grids using a combination of Weighted Least Squares (WLS) and an Adaptive Interpolation Extended Kalman Filter (AIEKF). This approach addresses the vulnerabilities in traditional state estimation methods by introducing an adaptive interpolation strategy that enhances the accuracy of the Kalman Filter in the presence of nonlinear data. The method was validated through extensive simulations on the IEEE-14-bus power system, where it effectively detected FDIAs by measuring the Euclidean distance between state estimations. The results demonstrated that this combined method significantly improves the detection capabilities and accuracy of the power system state estimation, making it a robust solution for maintaining grid security against sophisticated cyber-attacks. [6]

Chapter 3

SYSTEM OVERVIEW

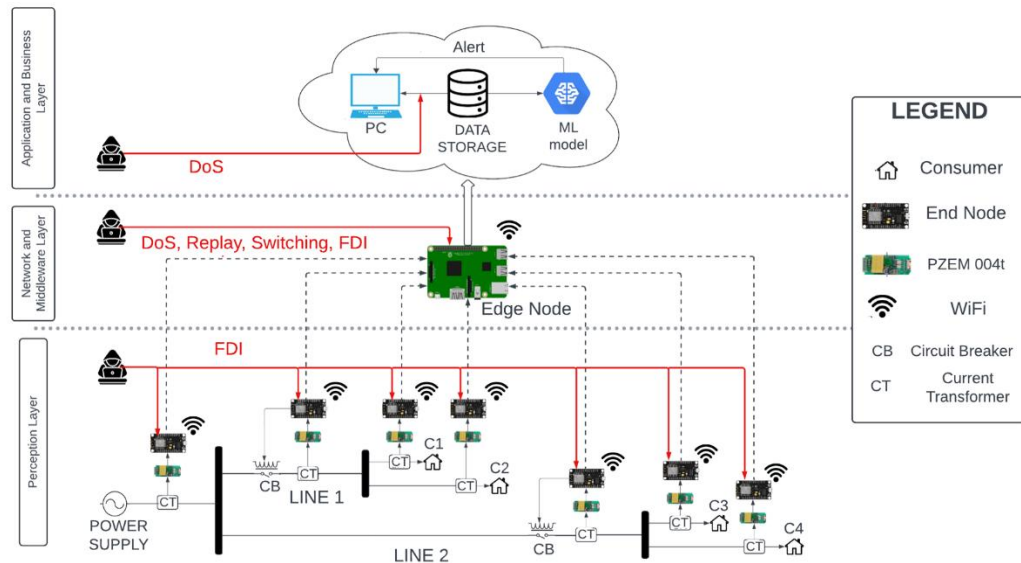


Fig. 01 System Block Diagram

The **System Block Diagram** shown above in Fig. 1 provides a visual representation of the entire "Real-Time Cyber Attack Detection in Power Systems" architecture, illustrating the interactions between hardware components, communication protocols, and the data processing infrastructure. The key components and layers are,

Perception Layer (Bottom Layer):

Components:

- Power supply sources connected to circuit breakers (CB).
- Current transformers (CT) attached to the electrical lines to measure power flow.
- End nodes (such as microcontrollers) associated with the PZEM 004t, a power meter, gathering data from consumers.
- The end nodes communicate wirelessly to the edge node, transmitting information about electrical parameters (voltage, current, power).

Purpose:

This layer monitors and measures the physical electrical parameters (energy consumption, voltage, current) in real-time from various consumer points (C1, C2, C3, C4) on two different electrical lines (Line 1 and Line 2).

Network and Middleware Layer:**Components:**

- Edge node (could be a Raspberry Pi or similar device).
- Wireless communication network (WiFi symbol), ensuring data exchange between end nodes and the edge node.

Function:

This layer collects and forwards the information from the perception layer's end nodes to the data storage or processing layer. The edge node aggregates data from all the end nodes and relays it to higher levels of the system for further processing and decision-making.

Cybersecurity Threats:

- Denial of Service (DoS)
- Replay attacks
- False Data Injection (FDI)
- Network Switching Attacks

Application and Business Layer:**Components:**

- Data storage system and machine learning (ML) models for real-time analytics.
- PC or monitoring systems, which generate alerts and process data.

Purpose:

This layer is responsible for analyzing the data gathered from the edge nodes using ML algorithms, generating alerts, and providing decision support for the grid operators. The data here is used to manage power distribution and detect any anomalies or attacks.

Cybersecurity Threats:

- Denial of Service (DoS) attacks could also occur in this layer, disrupting communication between the edge nodes and central storage or data processing systems.

3.1 METHODOLOGY

1. System Architecture Design:

- **Perception Layer Setup:**
 - Deploy **end nodes** (such as microcontrollers or ESP modules) at various points in the grid (C1, C2, C3, C4), with each node connected to a **PZEM 004t** power meter and **Current Transformers (CT)** for real-time monitoring of electrical parameters (voltage, current, power).
 - Install **Circuit Breakers (CB)** and **power supply** units in the physical layer, ensuring that the current transformers accurately measure the energy flow in both **Line 1** and **Line 2**.
 - Ensure wireless communication capabilities using WiFi to transmit data from each end node to the edge node.
- **Network and Middleware Layer:**
 - Set up an **Edge Node** (Raspberry Pi or similar device) to aggregate data from the end nodes.
 - Implement WiFi connectivity to ensure reliable data transmission from end nodes to the edge node.
 - Ensure proper data routing to allow real-time information transfer to the **data storage and ML model** for further analysis.
- **Application and Business Layer:**
 - Establish a **data storage system** capable of handling the continuous stream of data from the edge node.
 - Develop a **machine learning (ML) model** for analyzing the received data to detect anomalies, predict future usage, and generate alerts.
 - Implement a **monitoring system (PC)** that allows grid operators to receive alerts, visualize data, and make decisions.

2. Cybersecurity Threat Identification and Simulation:

- **Denial of Service (DoS) Attacks:**
 - Simulate and test DoS attacks in the network and middleware layer by flooding the edge node with excessive traffic to observe its ability to handle disruptions and to test system resilience.
- **False Data Injection (FDI) Attacks:**
 - Simulate FDI attacks by injecting corrupted or misleading data into the perception layer through end nodes to evaluate how the edge node and machine learning model detect and manage these anomalies.
- **Replay Attacks:**
 - Simulate replay attacks by sending previously recorded legitimate data back to the system to assess how the system differentiates between real-time data and replayed data.
- **Switching Attacks:**
 - Test switching attacks, where an adversary manipulates switching logic or interrupts connections in the network layer, affecting how data flows between the end nodes and the edge node.

3. Data Collection and Aggregation:

- Gather data from the **end nodes** via wireless transmission. Ensure that data such as voltage, current, and power are being accurately sent to the **edge node**.
- Validate that data aggregation occurs without loss or significant delay.
- Implement checks to verify that no **FDI attacks** or **replay attacks** are compromising the integrity of the data before it reaches the edge node.

4. Anomaly Detection with Machine Learning:

- Utilize the **collected data** to train a **machine learning (ML) model** for anomaly detection, focusing on identifying unusual patterns that could indicate cyberattacks or grid instability.
 - Example: Use supervised learning models (e.g., Random Forest, SVM) to classify incoming data as "normal" or "anomalous".

- Apply unsupervised learning methods (e.g., clustering) to detect novel attacks or faults.
- Continuously update the model with new data to improve its detection capabilities and robustness against novel threats.

5. Real-time Monitoring and Alerting:

- Implement real-time data processing capabilities to ensure that the **edge node** transmits the processed data to the **centralized storage and monitoring system**.
- Generate **alerts** in case of anomalies detected by the machine learning model (e.g., data inconsistency due to FDI, or network congestion due to DoS).
- Provide a **dashboard** where operators can view the current state of the grid, including energy usage, status of consumers, and any detected threats or issues.

6. Attack Mitigation Mechanisms:

- **DoS Mitigation:**
 - Deploy rate-limiting and filtering at the **edge node** to block traffic spikes and mitigate DoS attacks.
- **FDI Detection and Correction:**
 - Use the **ML model** to identify data that does not conform to normal patterns. Implement algorithms to discard or flag the anomalous data.
- **Replay Attack Prevention:**
 - Integrate timestamping or session identifiers in the data packets to detect and reject replayed data.
- **Switching Attack Response:**
 - Incorporate secure communication protocols between nodes to prevent unauthorized switching of devices or network rerouting.

7. System Validation and Testing:

- Test the system with multiple scenarios, including normal operation and under cyber-attacks (e.g., DoS, FDI, Replay attacks).
- Assess the system's performance by measuring:

- Data accuracy and integrity before and after cyber-attack simulations.
- The speed and accuracy of anomaly detection by the machine learning model.
- System resilience and recovery time after cyber-attacks (e.g., DoS).

8. System Deployment and Maintenance:

- After validation, deploy the system in the live environment.
- Establish a regular **monitoring and maintenance routine** to update the machine learning models with new data and enhance detection algorithms based on the latest threat intelligence.
- Regularly test the system for vulnerabilities and ensure that all communication protocols are up-to-date with security patches.

3.2 COMPONENTS:

Table I. Required Components

S. No	Components	Requirements
1	NodeMCU	7
2	PZEM 004t	7
3	Raspberry Pi	1
4	USB Adapter 2dBi Antenna	1
5	Resistive Loads	4
6	Power Supply	Req. Qty
7	Connecting Wires	Req. Qty

Table I presents a comprehensive list of the hardware and software components essential for implementing the power system network, data acquisition, and real-time monitoring for cyber attack detection. Each component plays a critical role in establishing the power grid setup, communication infrastructure, and data processing required to achieve the project's objectives.

Chapter 4

WORK PROGRESS

The project has made significant strides in the development and implementation of a real-time cyber-attack detection system for power grids. The work progress can be broken down into several key milestones, reflecting the systematic approach taken to integrate various components and achieve a functional prototype.

4.1 Establishment of Power System Network

1. Physical Layer Setup:

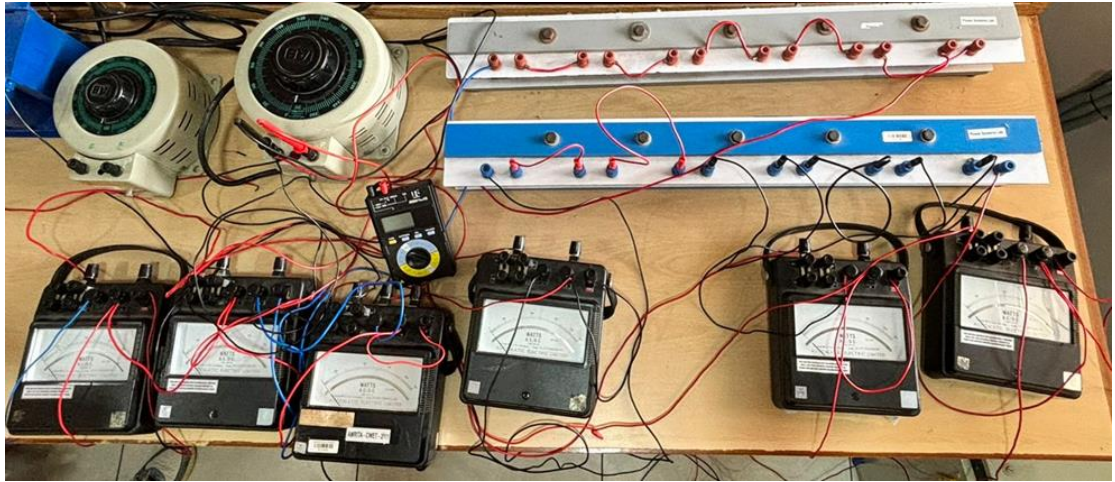


Fig.2 Physical Setup of Power System Network

The first phase of the project involved setting up a small-scale power system network. The setup shown in Fig.2 included essential components such as auto transformers, transmission lines (simulating power distribution), resistive loads, wattmeters, and multimeters. This network provided a realistic environment to test the cyber-attack detection system under controlled conditions. The results are tabulated in Table II below.

Table II. Output Readings from the Experiment Performed

S.No	Source Voltage (V)	Source Power (W)	Tr1 (W)	Tr2 (W)	L1 (W)	L2 (W)	L3 (W)
1	224	1240	120	1080	120	360	680
2	222	1520	240	1200	240	480	680
3	219	2120	720	1360	680	720	640
4	218	2400	720	1640	680	1000	620

2. Sensor Integration:

PZEM004t energy meters were integrated into the network to measure electrical parameters including voltage, current, power, energy consumption, frequency, and power factor. This data forms the basis for monitoring the health and stability of the power system.

4.2 Integration of NodeMCU with PZEM004t Sensors

1. Sensor-NodeMCU Interface:

NodeMCU modules were configured to interface with the PZEM004t sensors. The communication between the sensors and the NodeMCU was established using the UART protocol. This setup shown in Fig.3 allowed the NodeMCU to read real-time data from the sensors and process it locally.

2. Programming and Calibration:

The NodeMCU modules were programmed using the Arduino IDE with the <PZEM004TV30.h> library, which facilitated the communication with the PZEM004t sensors. Calibration routines were performed to ensure accurate readings from the sensors under different load conditions. This step was crucial for validating the system's ability to collect reliable data.

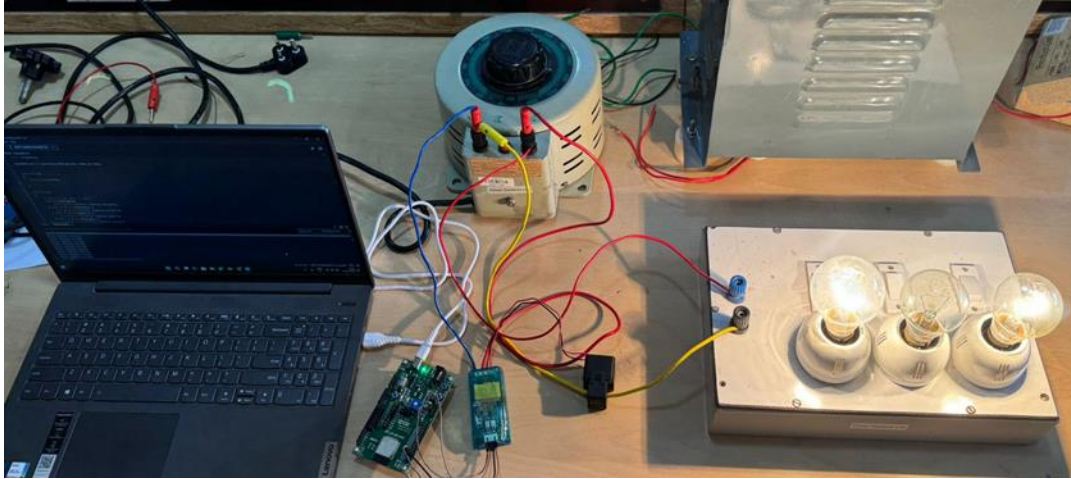


Fig.3 Sensor Connection with NodeMCU

3. Real-Time Data Acquisition:

After successful calibration, the NodeMCU modules were able to continuously acquire data from the PZEM004t sensors. The data included vital metrics such as voltage, current, active power, energy consumption, frequency, and power factor. This data was then transmitted to the Raspberry Pi server for further processing.

4.3 Development of a Centralized Server using Raspberry Pi

1. Server Configuration:

A Raspberry Pi was configured as a centralized server as shown in Fig. 4 to collect and manage data from multiple NodeMCU modules. The server setup included installing the necessary software libraries and frameworks for data handling and communication. The Raspberry Pi served as the main processing unit for aggregating data from the power system.

2. Communication Network Establishment:

A robust communication network was established between the NodeMCU modules and the Raspberry Pi server using the MQTT protocol over Wi-Fi. Each NodeMCU was configured to act as a client, publishing data to specific topics on the MQTT broker running on the Raspberry Pi. This setup enabled efficient and reliable data transfer from the sensor nodes to the server.

3. Data Handling and Storage:

The Raspberry Pi was equipped with a lightweight database to store the incoming data. The system was designed to handle high-frequency data inputs from multiple nodes, ensuring that no data points were lost during transmission. This step was critical for maintaining a complete dataset for analysis and training of machine learning models.

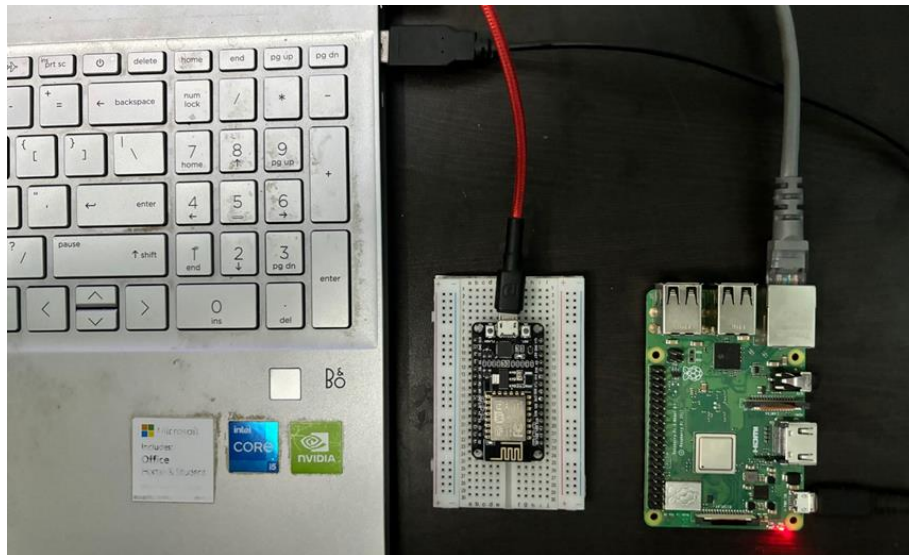


Fig.4 Raspberry Pi Configuration and Server Setup

Chapter 5

RESULT AND CONCLUSION

5.1 RESULTS

1. Successful Integration of Hardware Components

- The power system network was established, integrating CT & PT sensors, circuit breakers, relays, and loads for real-time monitoring.

2. Data Acquisition from NodeMCU and PZEM004t

- Accurate voltage, current, and power readings were collected using the PZEM004t sensor integrated with NodeMCU.
- Continuous data transmission was achieved through the MQTT protocol.

3. Raspberry Pi Configuration and Cloud Data Availability

- The Raspberry Pi successfully acted as an edge node, enabling real-time data transmission to the cloud.
- Sensor data was stored and made accessible on the server for visualization and further analysis.

5.2 CONCLUSION

- The power system network and its critical components were successfully set up, allowing real-time monitoring of electrical parameters.
- Integration of the NodeMCU with the PZEM004t sensors provided accurate and timely data on power consumption, which was effectively transmitted to the cloud using the Raspberry Pi.
- These foundational steps enable the next phase of the project—cyber attack simulations and machine learning-based anomaly detection, setting the stage for the development of a robust cyber security framework for power systems.

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